

FDT PE Factory Data Study

Dr. Scientist

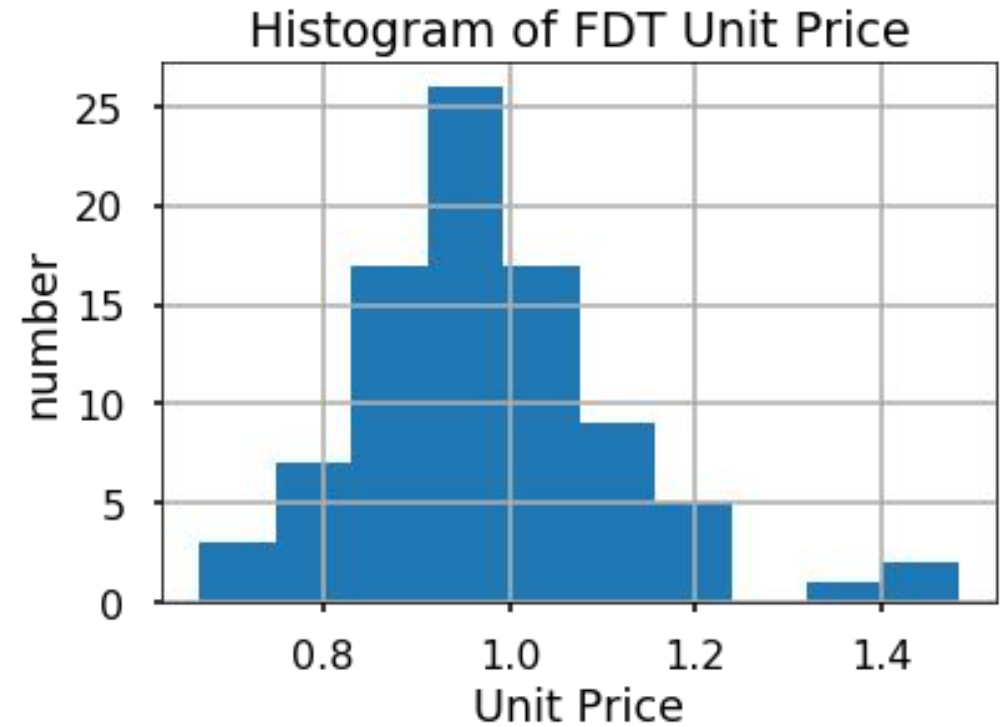
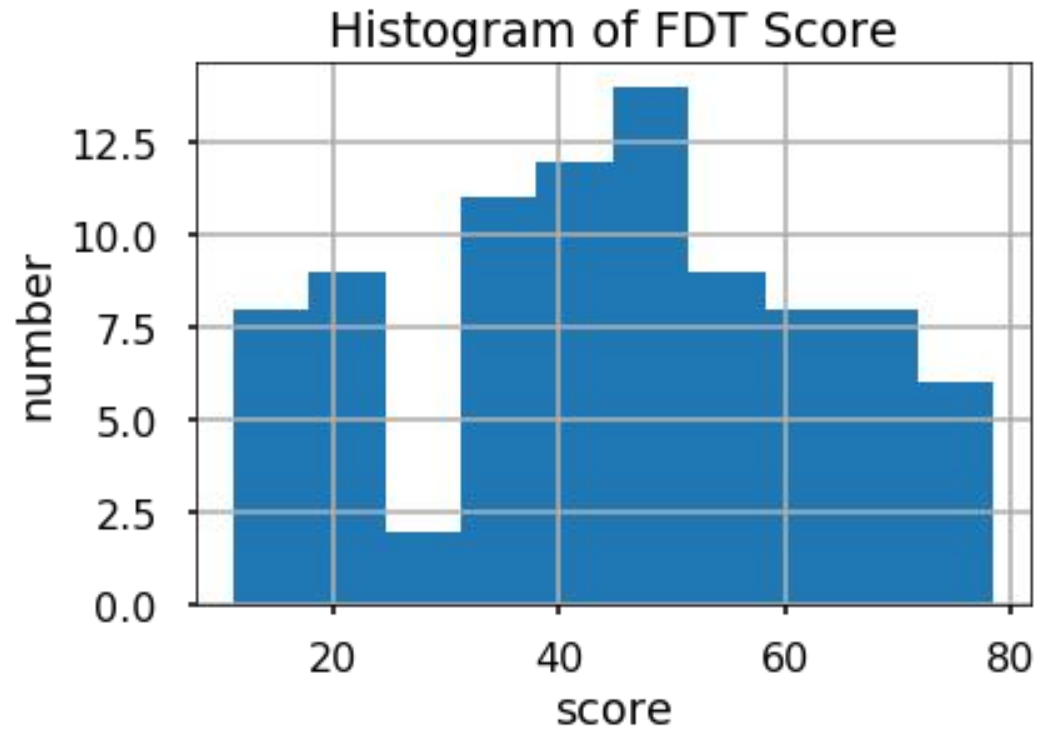
2017-Nov-22

Outline



- **PE Factory with FDTScore**
- Portfolio Strategies Details

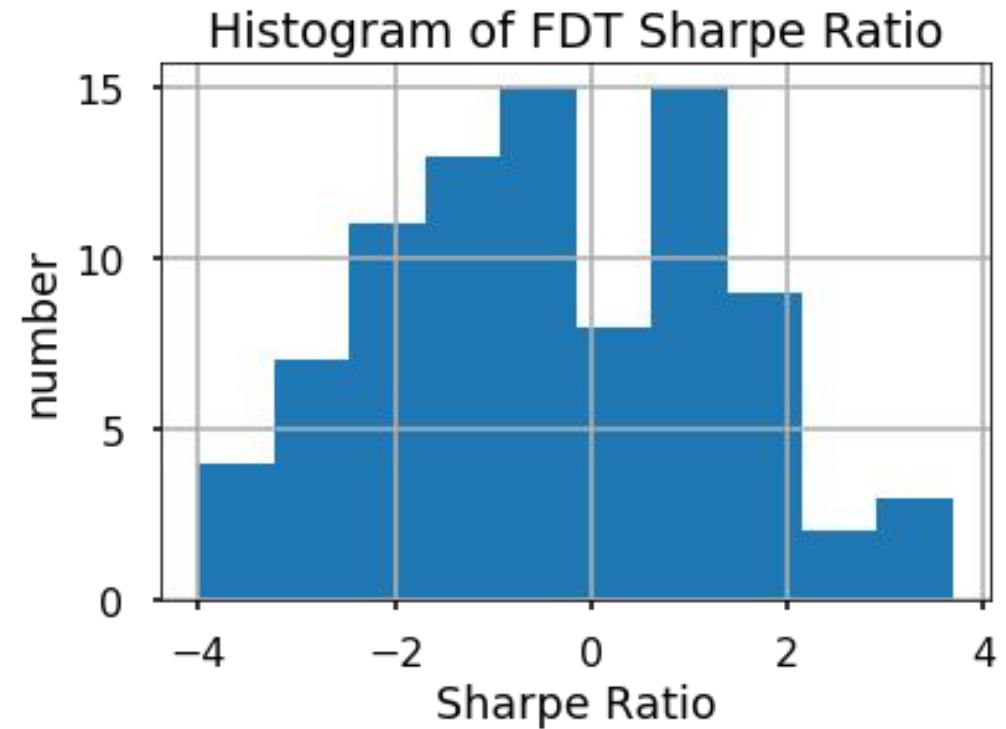
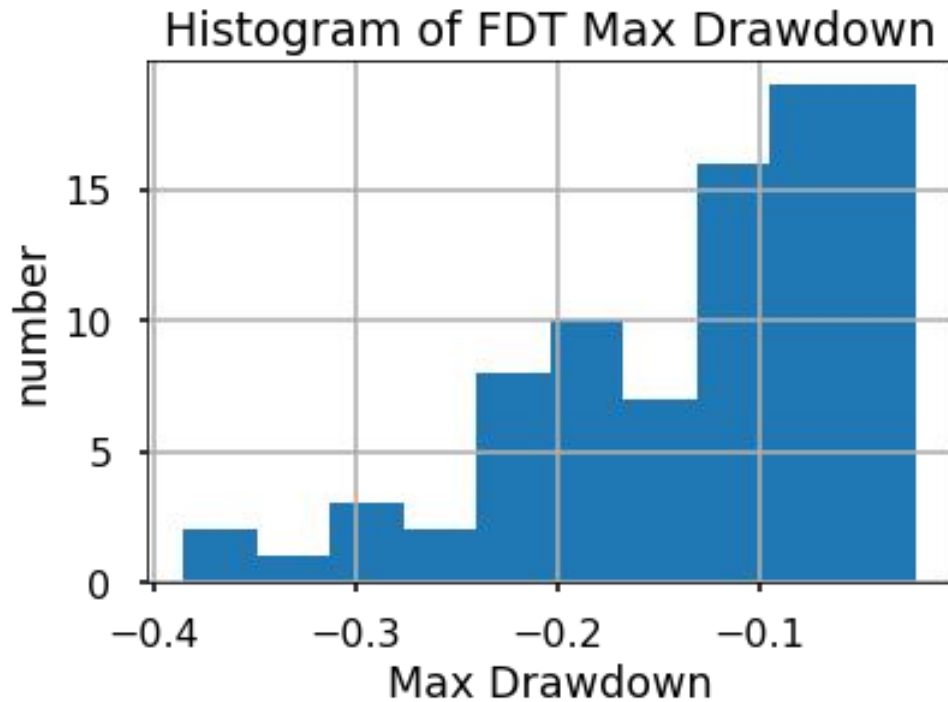
FDTScore Statistics



Overall PnL of all traders up to 2017-11-21

- USD: -107,127.00

FDTScore Statistics

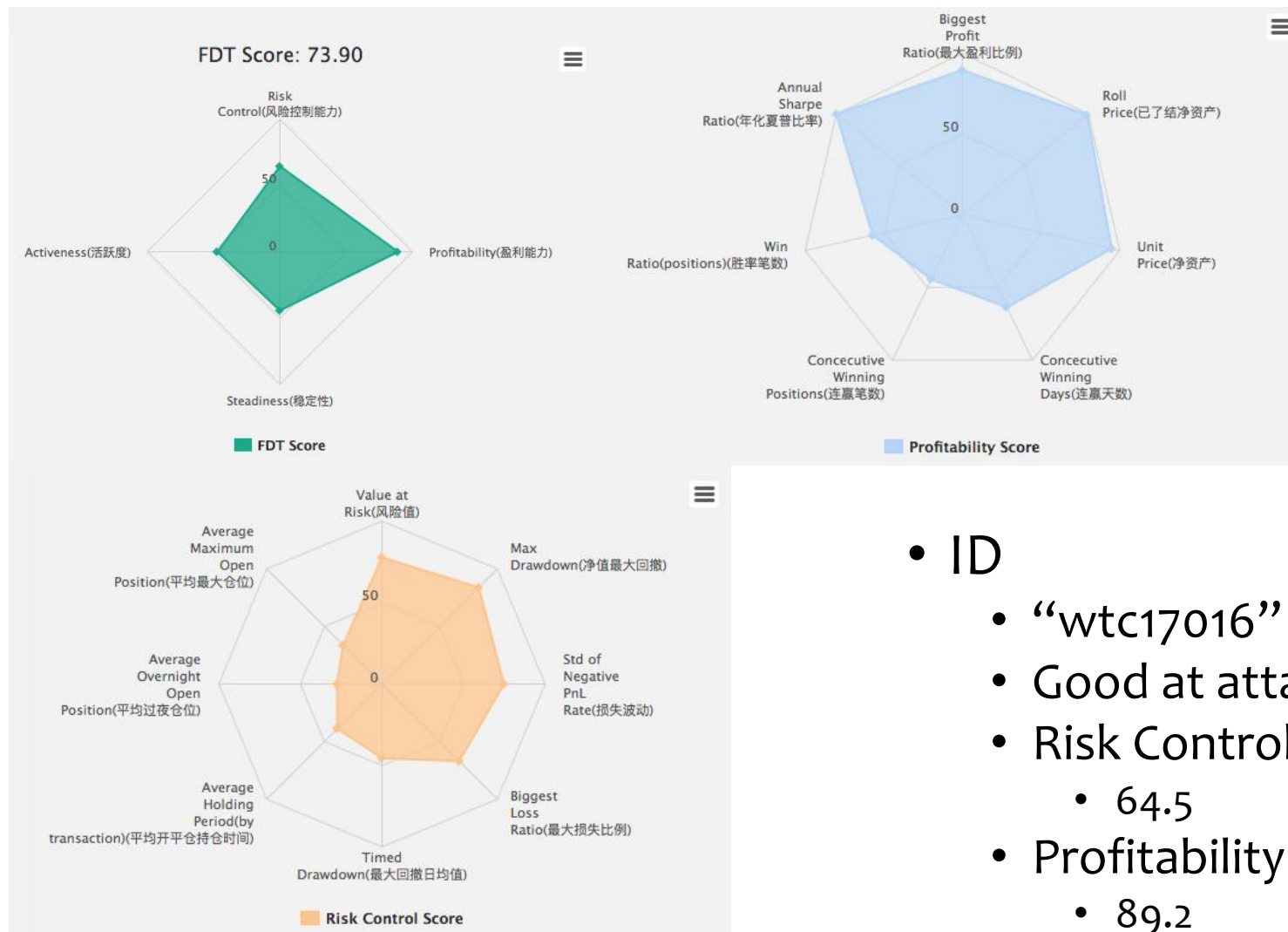


Talents thru FDTScore



- ID
 - “wtc17009”
 - Balanced skills
 - Risk Control
 - 80.6
 - Profitability
 - 80.5

Talents thru FDTScore



- ID
 - “wtc17016”
 - Good at attacking
 - Risk Control
 - 64.5
 - Profitability
 - 89.2

Outline

- PE Factory with FDTScore
- **Detail about Portfolio Strategies**
 - **Study Items**
 - **FDTScore: Diversity Analysis**
 - **FDTScore: Soft-unique Analysis**
 - **FDTScore: Hedging Effect Analysis**

Portfolio Strategy “beecu001”

U.S. Stock

FDTAI



• Portfolio “beecu001”

- Started on 2017-08-14 (68 days)
- Market: US Stock
- Benchmark: S&P 500
- Alpha: -0.02
- Beta: 1.05
- Max Drawdown: -3.3%
- Annual Ret.: 21%
- Annual Volatility: 8.9%

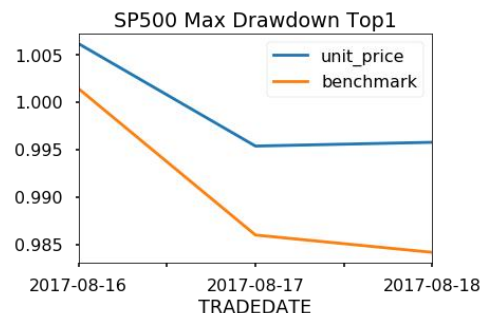
Portfolio Strategy “beecu001”

Blue: strategy

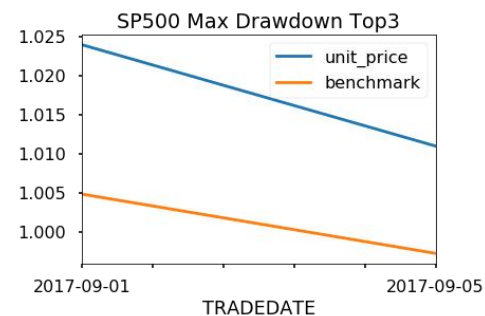
Orange: Benchmark

FDTAI

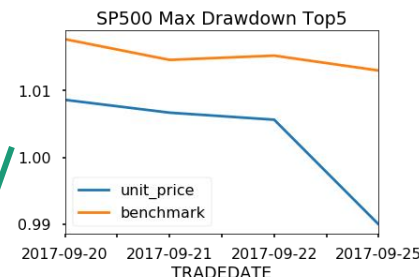
- Drawdown analysis



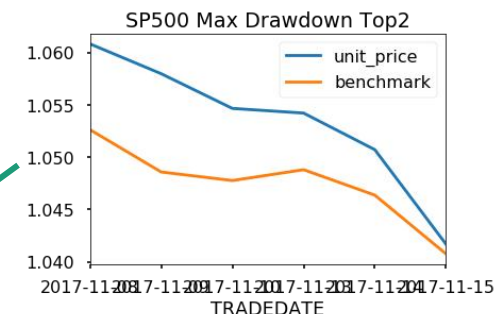
Top-1



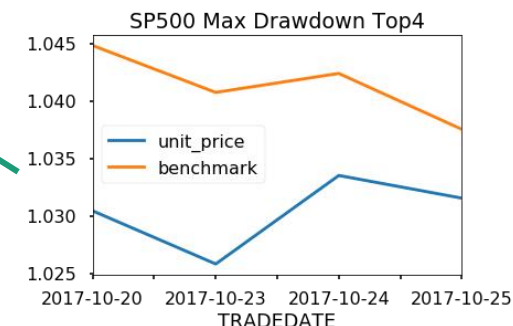
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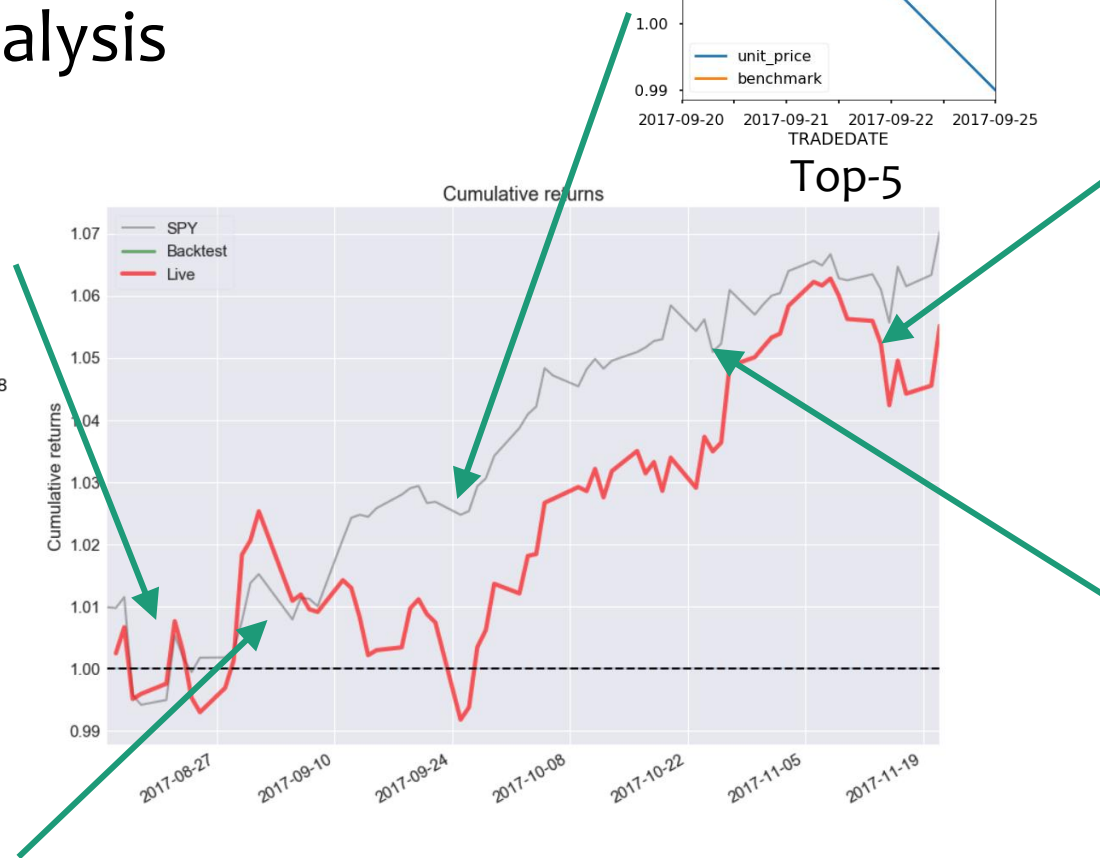
Top-5



Top-2



Top-4

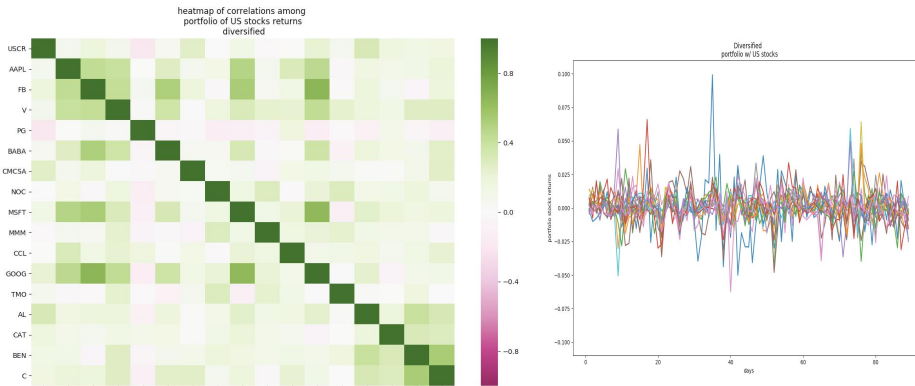


Diversity

1. How similar are your assets?
 - a. When you hold many assets, but many of your assets are similar, then you aren't really diversifying.
2. How many unique assets are you actually holding?
 - a. When you hold many assets, but many of them are similar, how many unique assets do you really have? We can quantify the exact number of unique assets you actually hold.
3. When it looks like you have diversified enough, have you really achieved the *goal of diversification - reducing risk*?
 - a. We can quantify the amount of risk decreased by your portfolio.

Diversity: beecu001 U.S. stocks

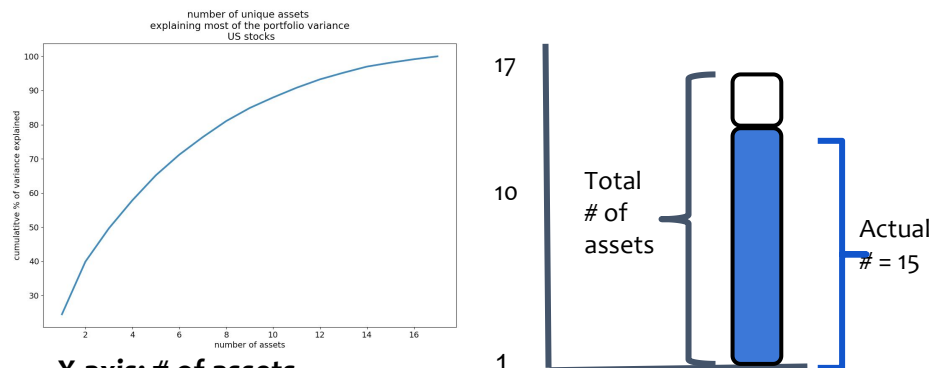
Fig. 1, 2: Portfolio asset similarity



X & Y axes: assets. Asset Correlations: **Green = similar;** **Light = different; Pink = hedge**

X-axis: days. Y-axis: asset returns $\in [-10\%, +10\%]$

Fig. 3, 4: Actual # of assets in portfolio



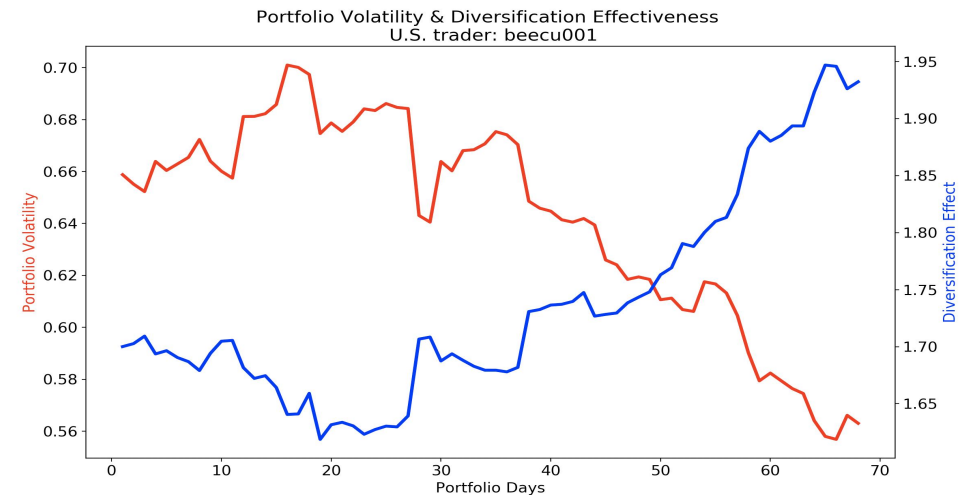
X-axis: # of assets. Y-axis: cumulative % of portfolio variance explained

The goal is to REDUCE RISK from diversification. Thus, the effect of diversification can be measured by this ratio - portfolio's weighted average asset volatility to its actual volatility. This measures the **essence of diversification, because we can quantify the amount of risk reduced by portfolio diversification!**

Since different asset classes are not perfectly correlated to each other, this ratio in general > 1 . In other words, the overall risk of a well-diversified portfolio is less than the weighted-average risk of its component holdings. $DE \in [1, \infty)$, where 1 indicates no diversification effect, larger number means better diversified. $DE = \{ \sum_j^S (W_j * \sigma_j) \} / \sigma_p$

Where σ_p = total portfolio risk = $\sqrt{W^T \text{Cov} W}$, Cov is the variance covariance matrix on daily asset returns, W is vector of weights
 σ_j = asset j risk = $\sqrt{\sum_i^n (j_i - \mu_j)^2 / n}$, j_i is return of asset j on day i

Fig. 5 Portfolio Diversity Risk Reduction Effect



X-axis: days. Red Y-axis: portfolio risk. Blue Y-axis: DE

Overall risk was reduced by 15%

Portfolio risk: 0.5 - 0.7 DE: 1.6-1.95

Portfolio Strategy “beecu002”

HK Stock

FDTAI

• Portfolio “beecu002”

- Started on 2017-08-14 (68 days)
- Market: HK Stock
- Benchmark: HS Index

- Alpha: 0.15
- Beta: 0.95
- Max Drawdown: -4.0%

- Annual Ret.: 58.4%
- Annual Volatility: 15.7%



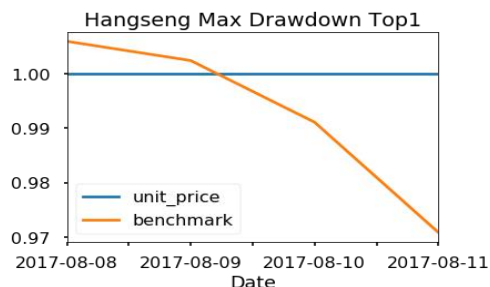
Portfolio Strategy “beecu002”

Blue: strategy

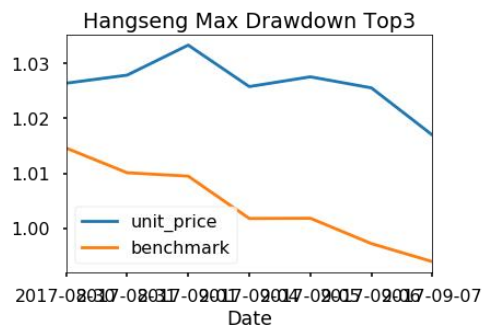
Orange: Benchmark

FDTAI

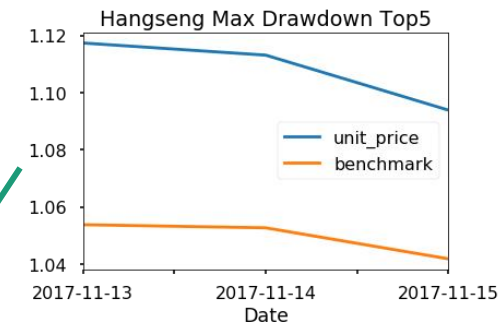
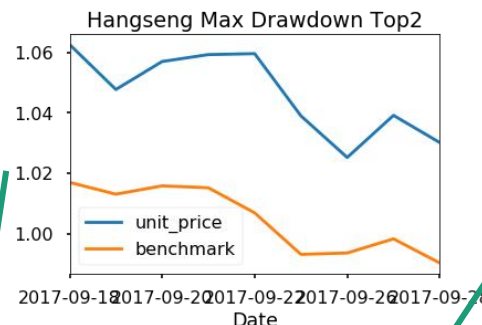
- Drawdown analysis



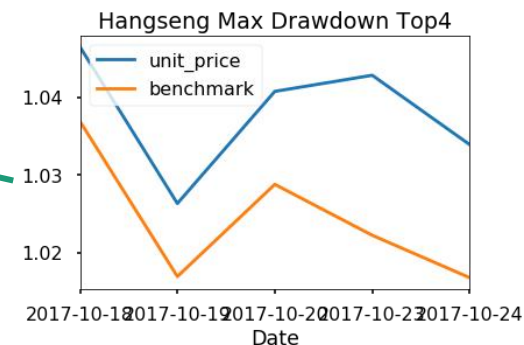
Top-1



Top-3



Top-5

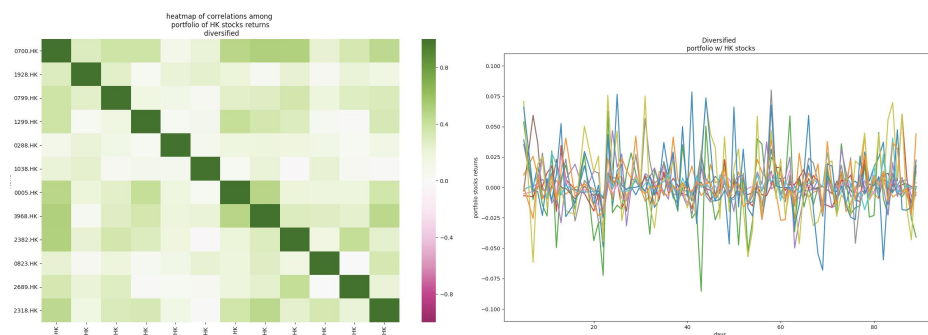


Top-4



Diversity: beecu002 HK stocks

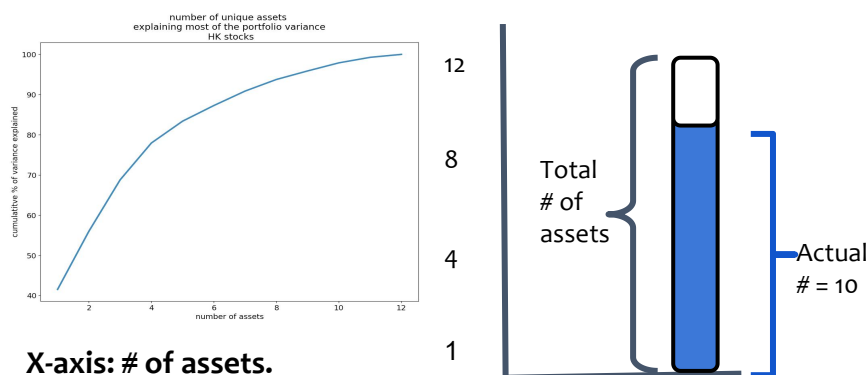
Fig. 1, 2: Portfolio asset similarity



X & Y axes: assets. Asset
Correlations: **green = similar;**
Light = different; Pink = hedge

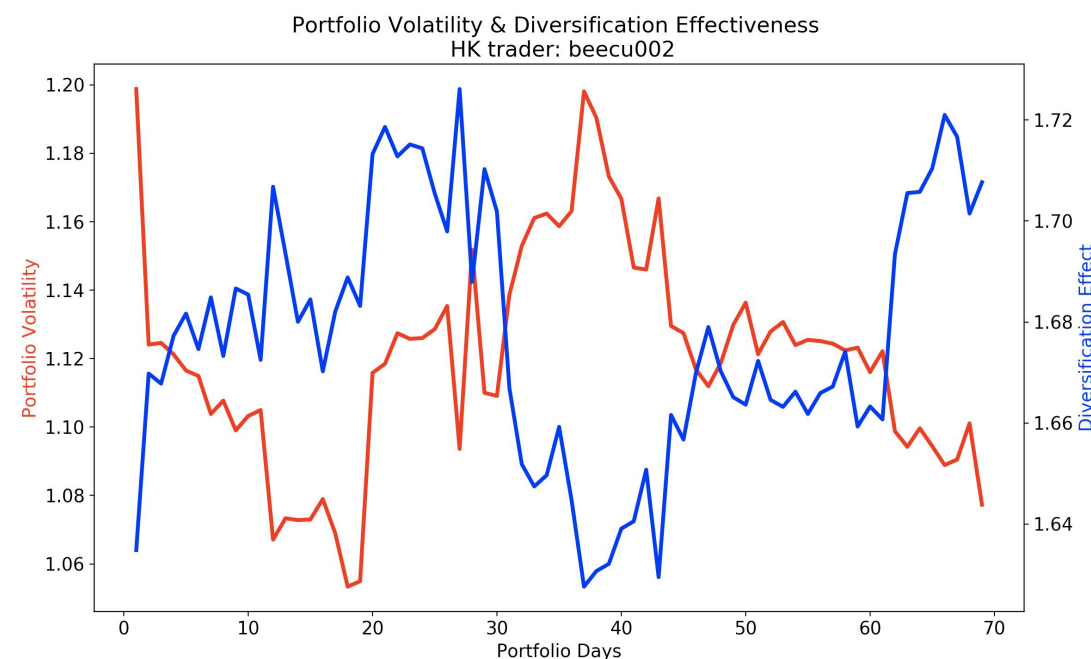
X-axis: days. Y-axis: asset
returns $\in [-10\%, +10\%]$

Fig. 3, 4: Actual # of assets in portfolio



X-axis: # of assets.
Y-axis: cumulative % of portfolio
variance explained

Fig. 5 Portfolio Diversity Risk Reduction Effect



X-axis: days. **Red Y-axis: portfolio risk. Blue Y-axis: DE**

Overall risk was reduced by 170%

Portfolio risk: 1 - 1.2 DE: 1.6 - 1.7

Portfolio Strategy “beecu003”

China A-share



• Portfolio “beecu003”

- Started on 2017-09-15 (43 days)
- Market: China-A
- Benchmark: HS300
- Alpha: 0.10
- Beta: 1.30
- Max Drawdown: -2.3%
- Annual Ret.: 133.7%
- Annual Volatility: 13.6%

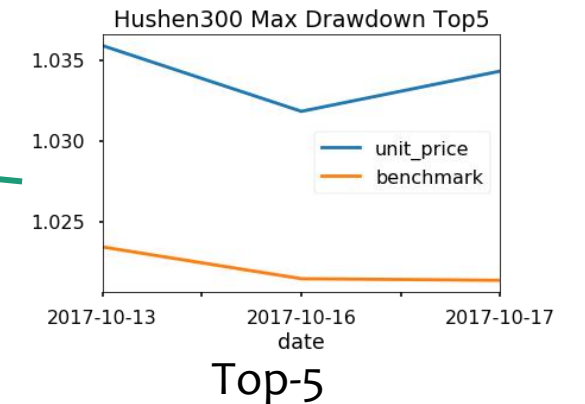
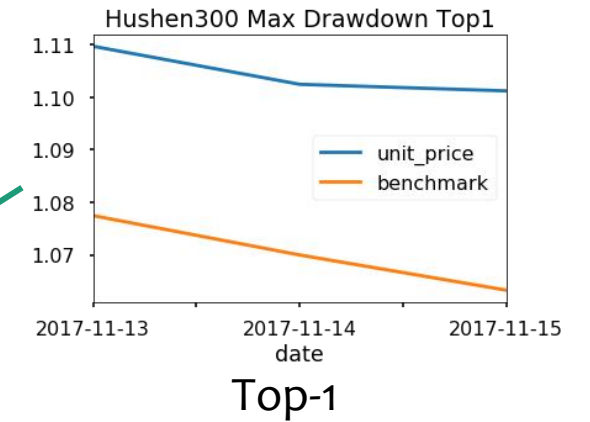
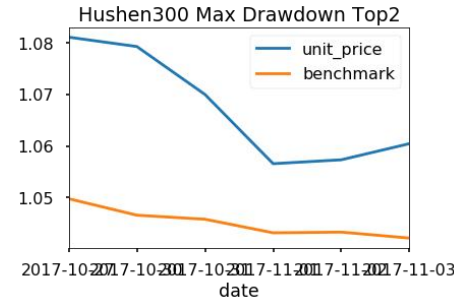
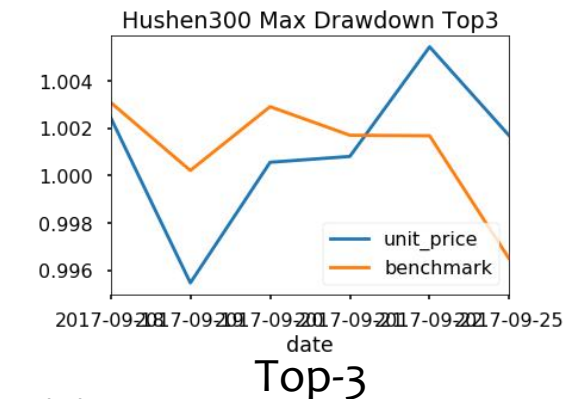
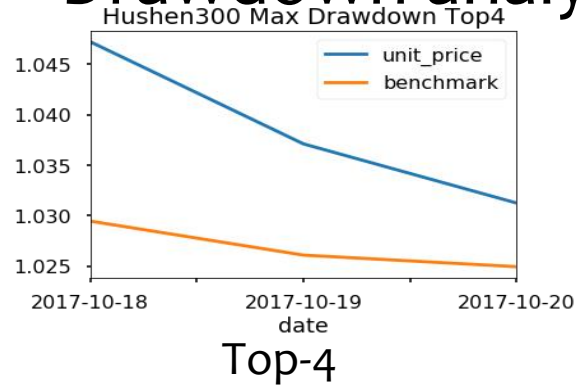
Portfolio Strategy “beecu003”

Blue: strategy

Orange: Benchmark

FDTAI

- Drawdown analysis



Diversity: beecu003 A stocks

Fig. 1, 2: Portfolio asset similarity

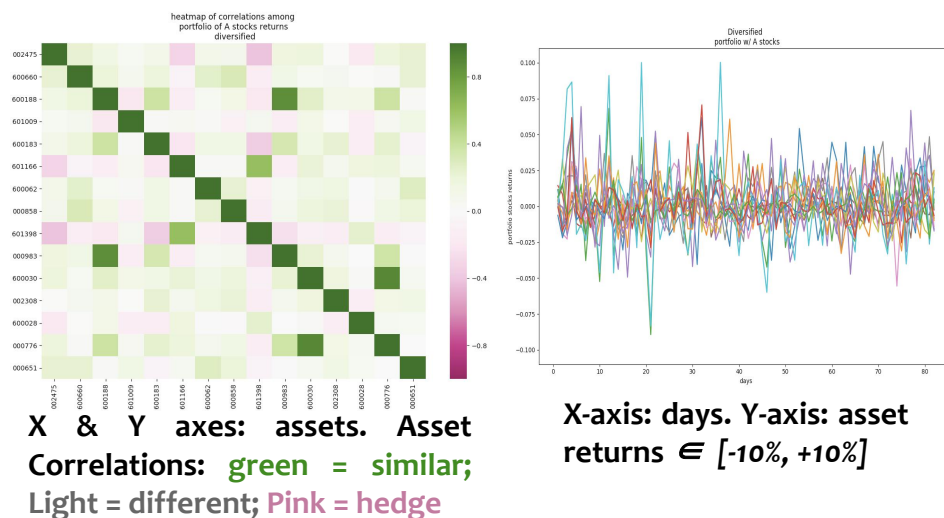
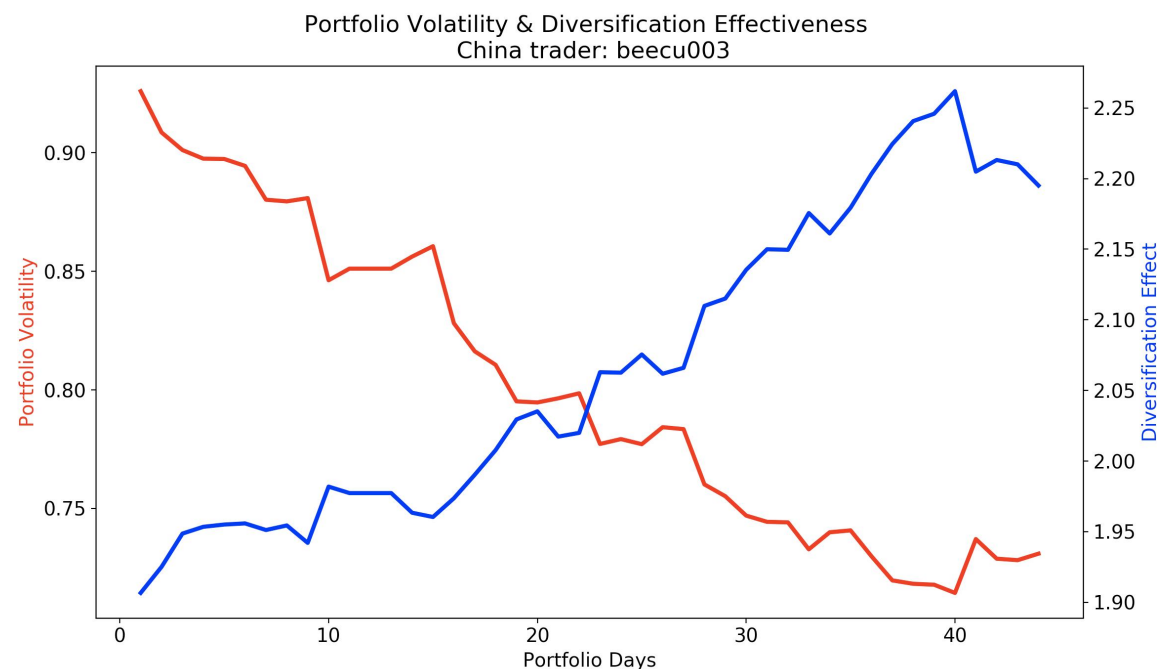
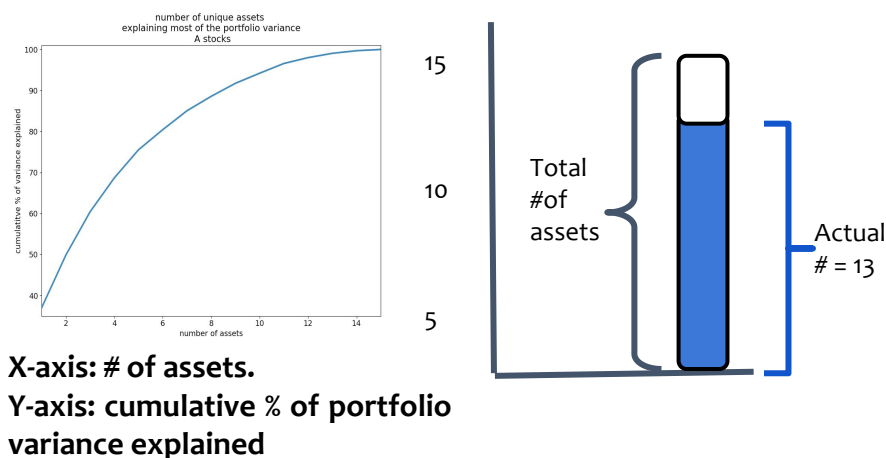


Fig. 5 Portfolio Diversity Risk Reduction Effect



X-axis: days. Red Y-axis: portfolio risk. Blue Y-axis: DE
Overall risk was reduced by 225%!
 Portfolio risk: 0.7 - 0.95 DE: 1.9 - 2.25

Fig. 3, 4: Actual # of assets in portfolio



Diversity Analysis

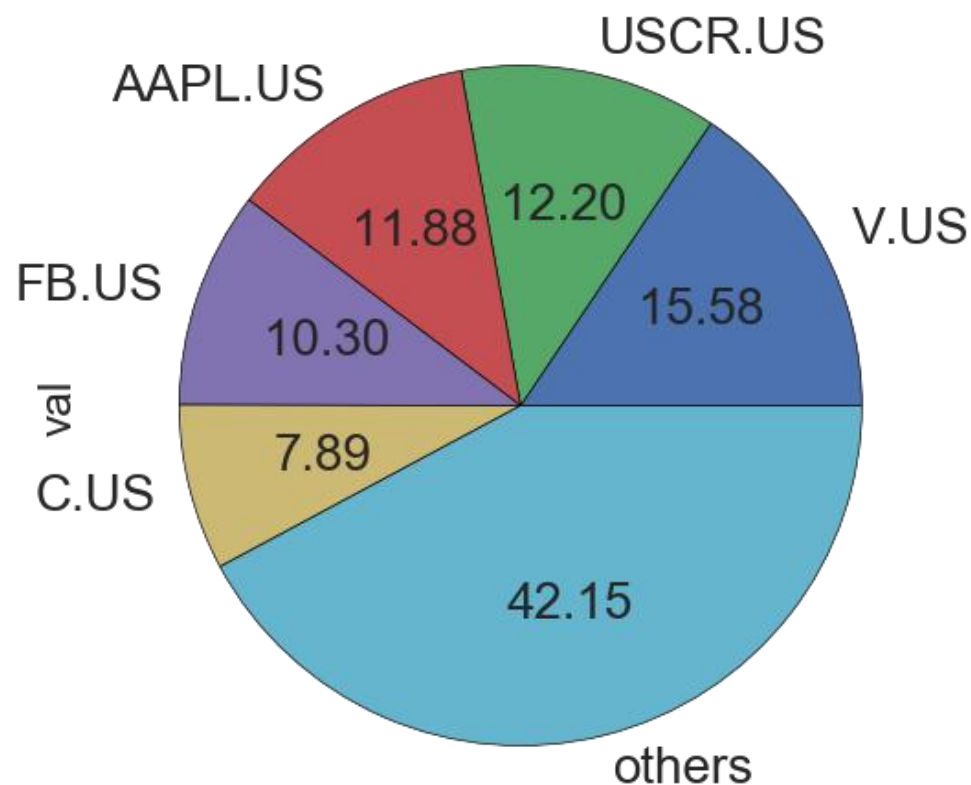
Asset similarity

- The trader in **A stocks** ('beecu003') market is the most-diversified, even showing strategies of hedging.
- The trader in **US stocks** ('beecu001') market is well-diversified, but not as good in A stocks market; there is less hedging.
- The trader in **HK stocks** ('beecu002') market is the least-diversified, compared to the other two. As shown by the heatmap, similarities are higher among the assets.

Diversity Effect in Risk Reduction

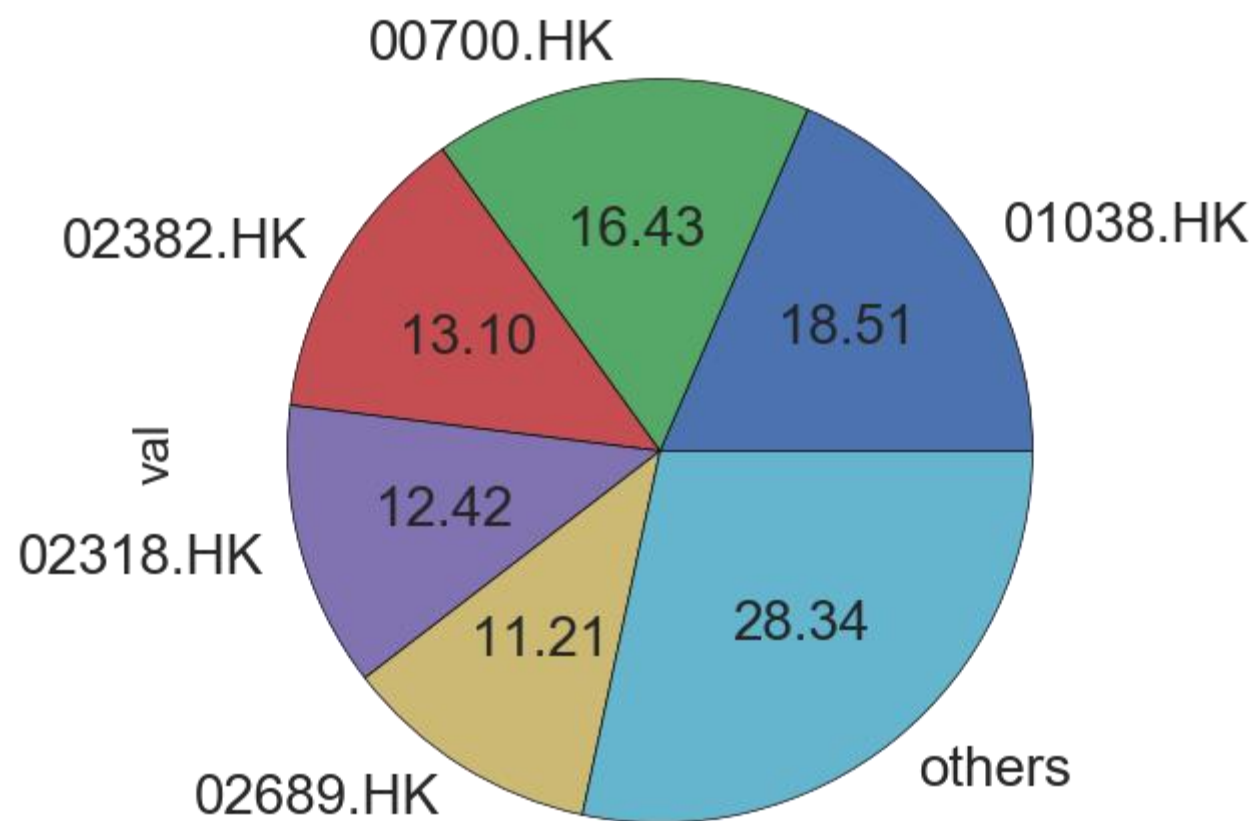
- beecu003 (A stocks): best trend - gradual and sustained improvement in DE. Portfolio is well adjusted for increasing diversification, and risk decreases over time.
- beecu001 (US stocks) also show this trend. **Their portfolio asset choices and weightings are chosen correctly over time.**
- beecu002 (HK stocks), DE more random and risk is volatile throughout. **Trader should to select his asset choices and weights better.**
- beecu003 in **A stocks** reduced risk *the most*, by 225%, and risk ranged from 0.70-0.95.
- beecu001 in **US stocks** reduced risk by 195%, and risk ranged from 0.56-0.70.
- beecu002 in **HK stocks** reduced risk *the least*, by 170%, and risk ranged from 1-1.2.
- This overall risk is lowest in US market and highest in HK market.

Heavy-position Assets – US Stock



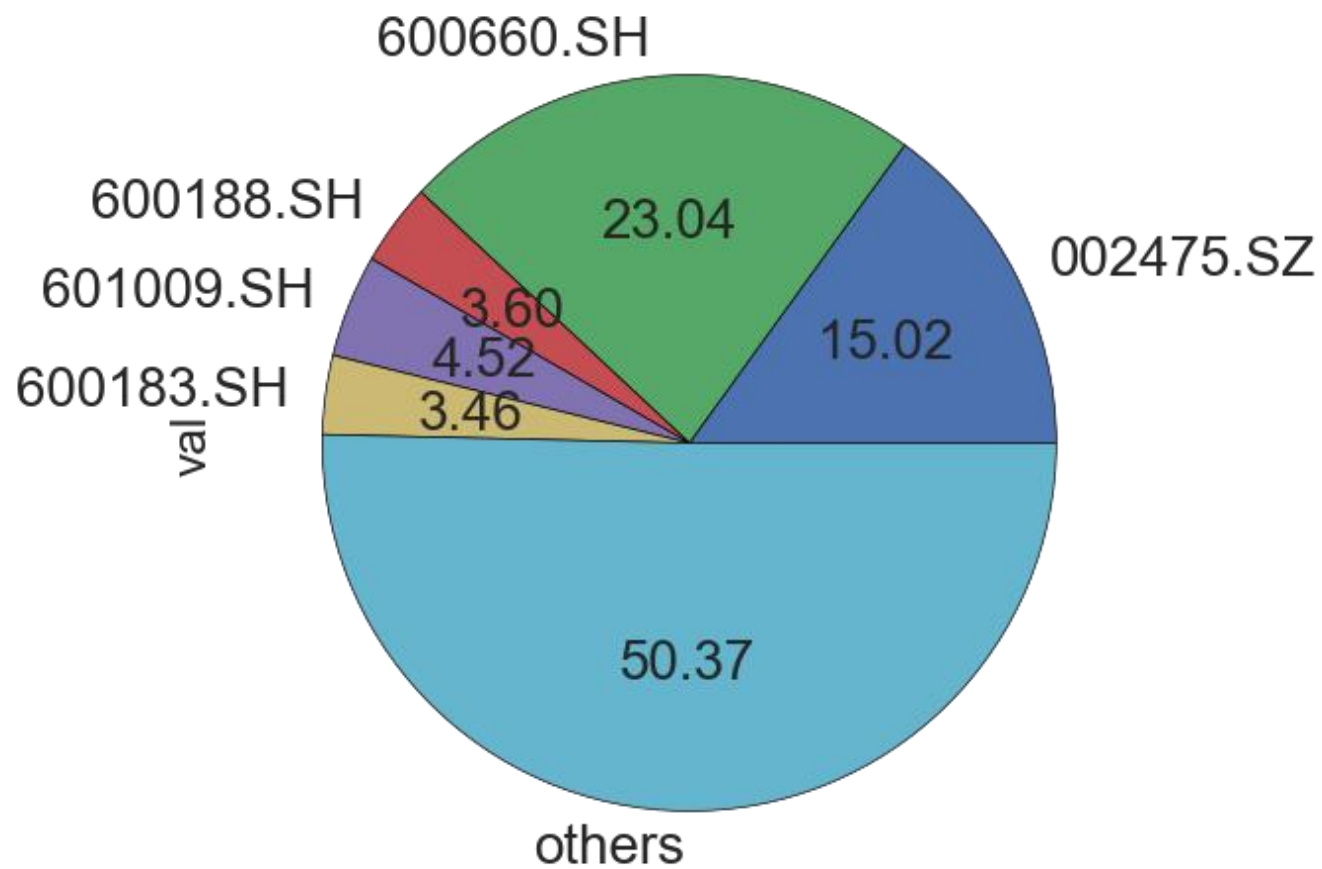
1	VISA
2	US CONCRETE
3	APPLE
4	FACEBOOK
5	CITIGROUP

Heavy-position Assets - HK Stock



1	長江基建集團
2	騰訊
3	舜宇光學科技
4	中國平安
5	玖龍紙業

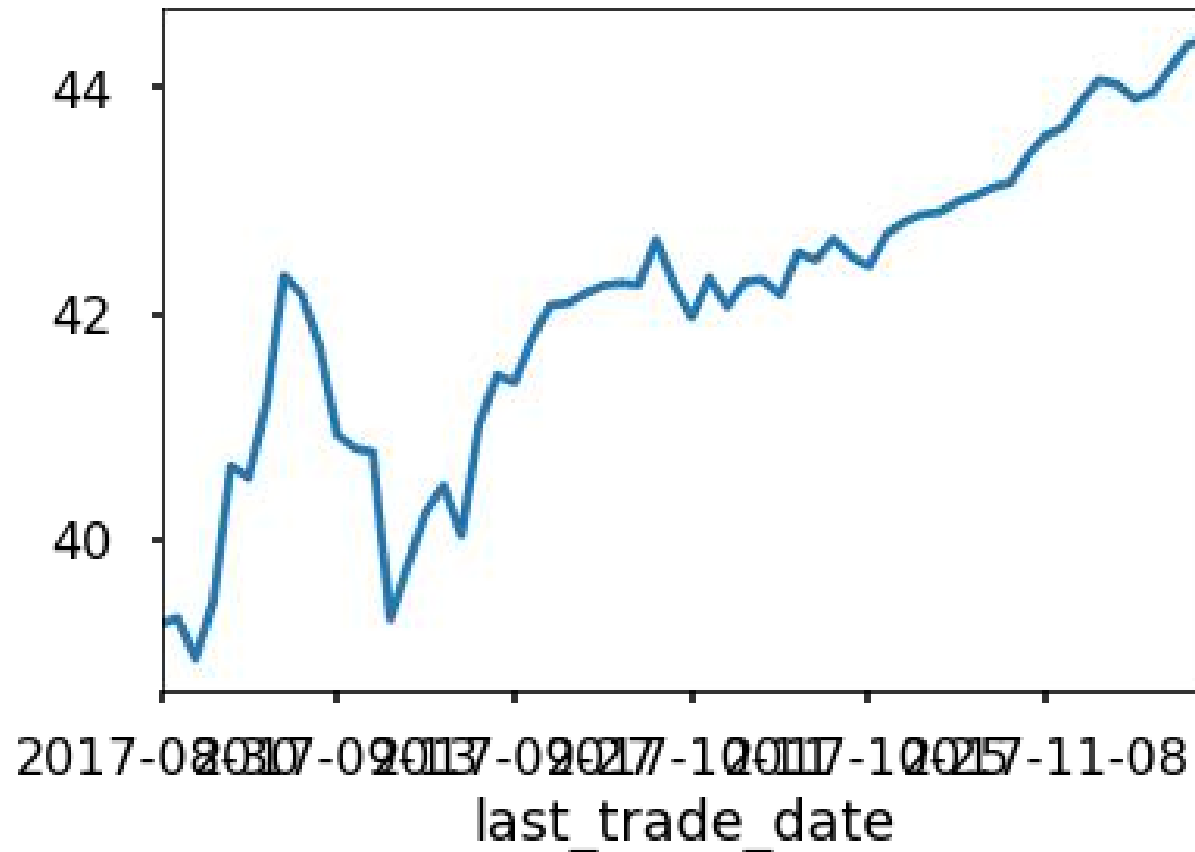
Heavy-position Assets – A share



1	立讯精密
2	福耀玻璃
3	兖州煤业
4	南京銀行
5	生益科技

Backup Slides

FDTScore Statistics



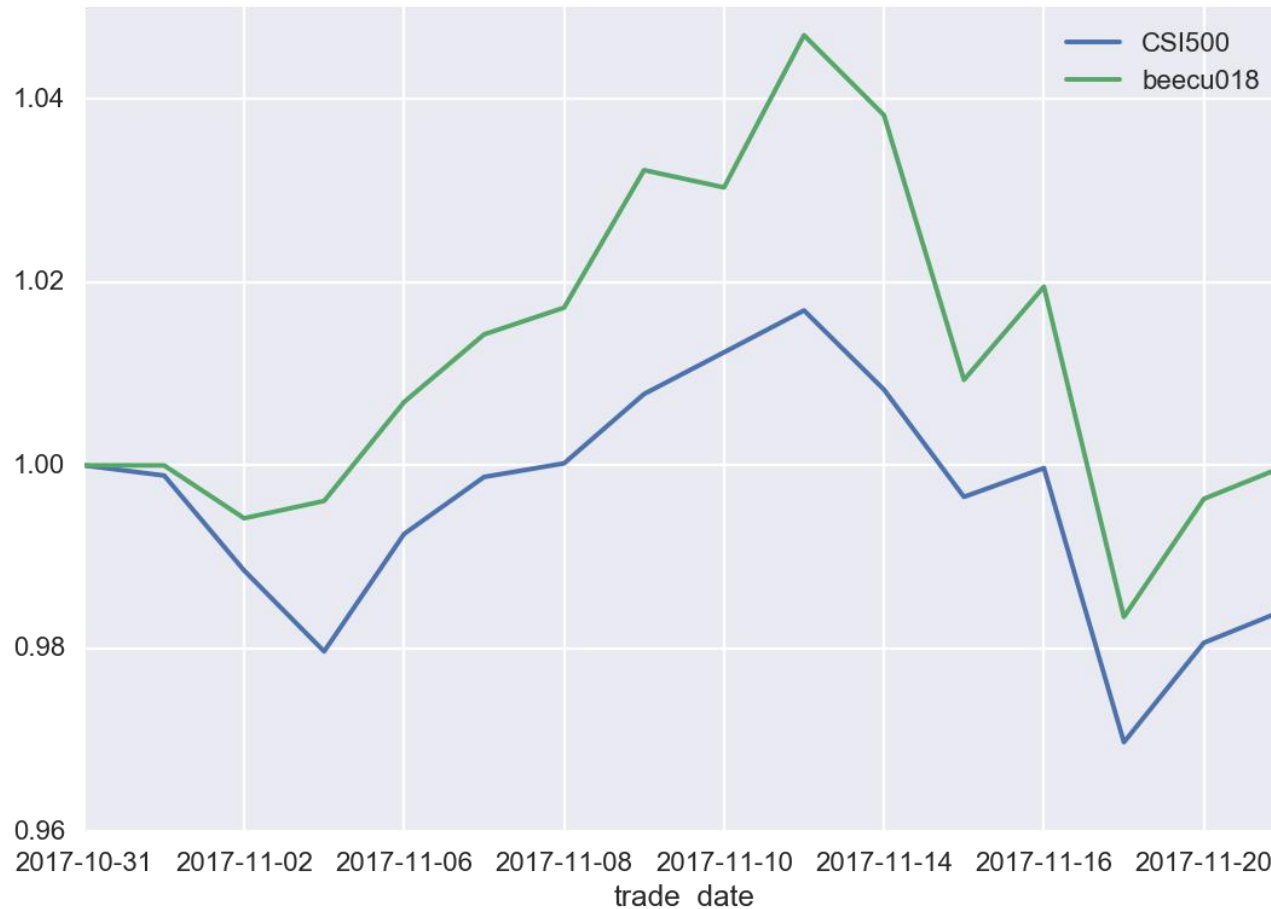
Portfolio Strategy “beecu008”



• Portfolio “beecu008”

- Started on 2017-10-31
- Market: China A
- Benchmark: HS300
- Alpha: -0.09
- Beta: 2.07
- Max Drawdown: -2.6%
- Annual Ret.: 281.8%
- Annual Volatility: 26.0%

Portfolio Strategy “beecu018”



• Portfolio “beecu018”

- Started on 2017-10-31
- Market: China A
- Benchmark: IC500
- Alpha: 0.39
- Beta: 0.98
- Max Drawdown: -6.1%
- Annual Ret.: -0.2%
- Annual Volatility: 22%

FDTScore: Diversity Analysis - "beecu001"

U.S. Stock

FDTAI

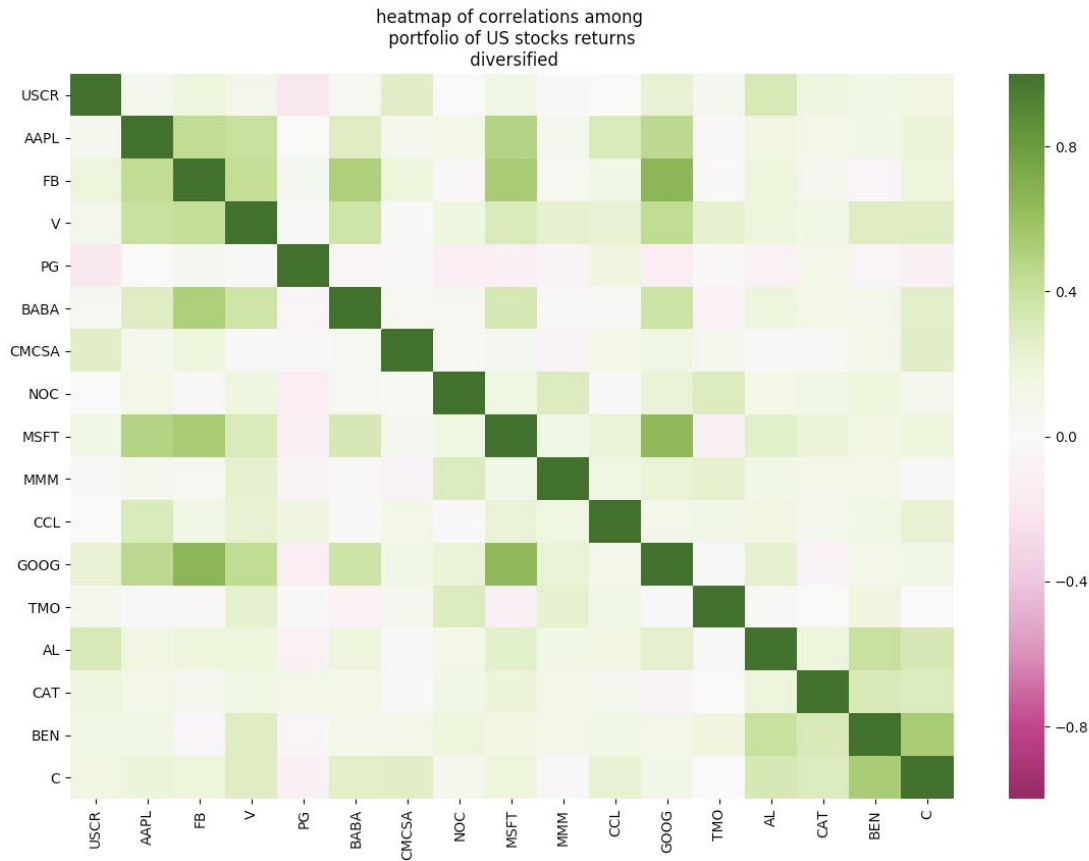
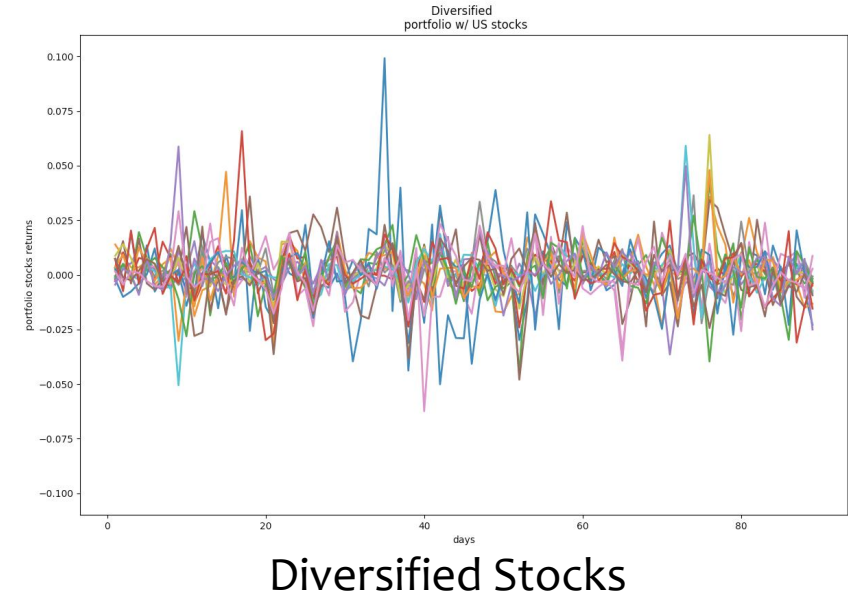


Fig. 1 Heatmap of correlations among portfolio assets' returns - Diversified (beecu001)

Portfolio Stocks Returns



- Remarks
 - Both plots shows the trader's portfolio being well diversified.

FDTScore: Soft-unique Analysis – ‘beecu001’^{U.S. Stock}



Fig. 3 Number of assets explaining portfolio variance

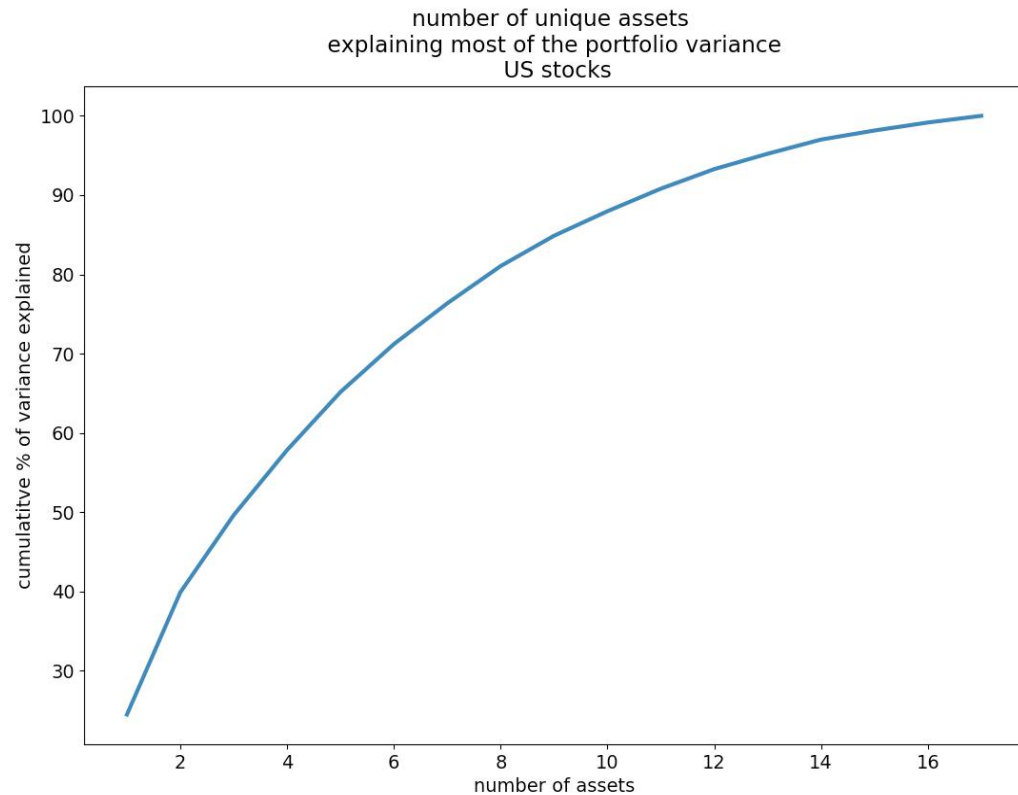
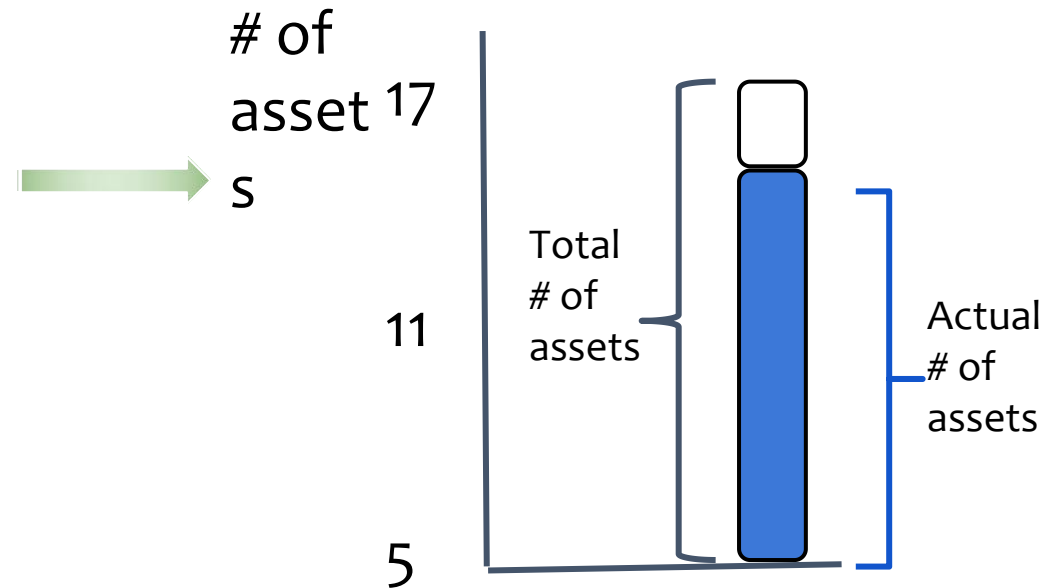


Fig. 4 Actual number of assets in portfolio

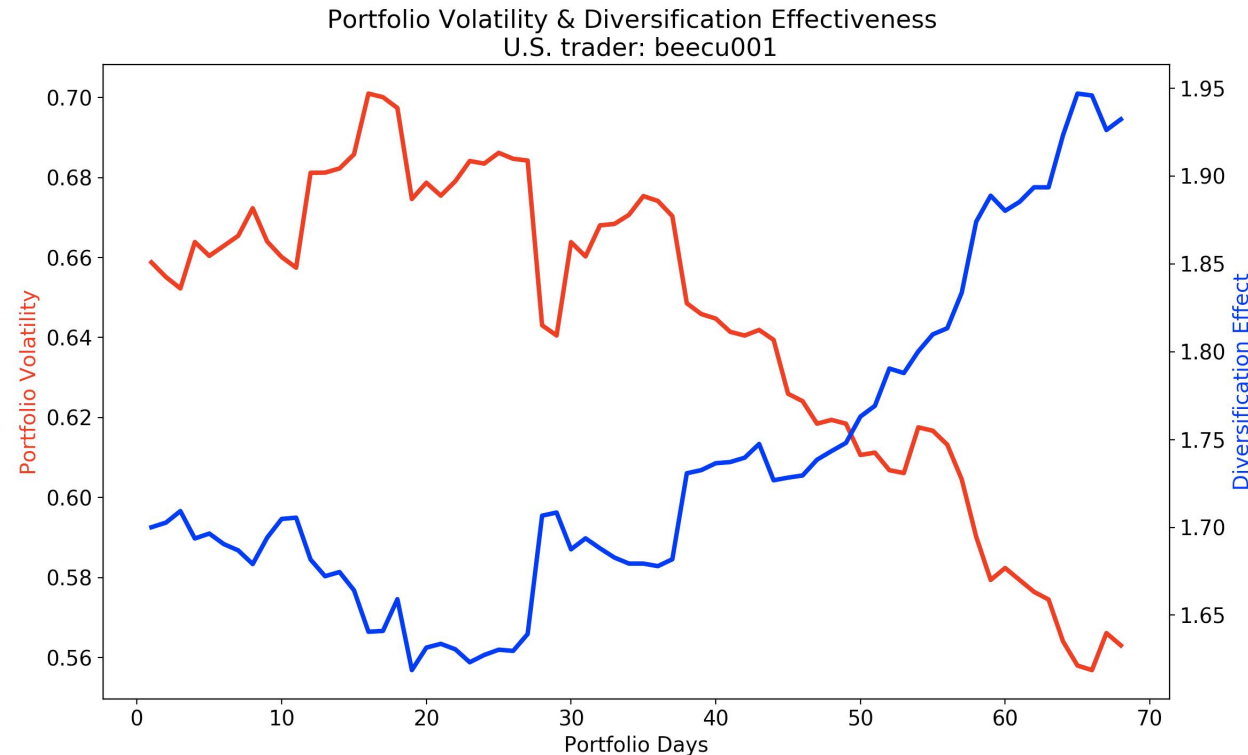


FDTScore: Hedging Effect Analysis – ‘beecu001’



The goal is to REDUCE RISK from diversification. Thus, the effect of diversification can be measured by this ratio - portfolio's weighted average asset volatility to its actual volatility. The result of this calculation measures the *essence of diversification*, *because we can quantify the amount of risk reduced by portfolio diversification!*

Since different asset classes are not perfectly correlated to each other, this ratio in general > 1 . In other words, the overall risk of a well-diversified portfolio is less than the weighted-average risk of its component holdings. $DE \in [1, \infty)$, where 1 indicates no diversification effect, larger number means better diversified.



Initially, when DE was lower ~ 1.7 , portfolio volatility was higher ~ 0.7 ; then the trader's diversifying effect increased in the latter half to 1.95, effectively reducing portfolio risk to 0.56. (overall portfolio risk was reduced by 195%!)

FDTScore: Diversity Analysis - "beecu002"

HK Stock

FDTAI

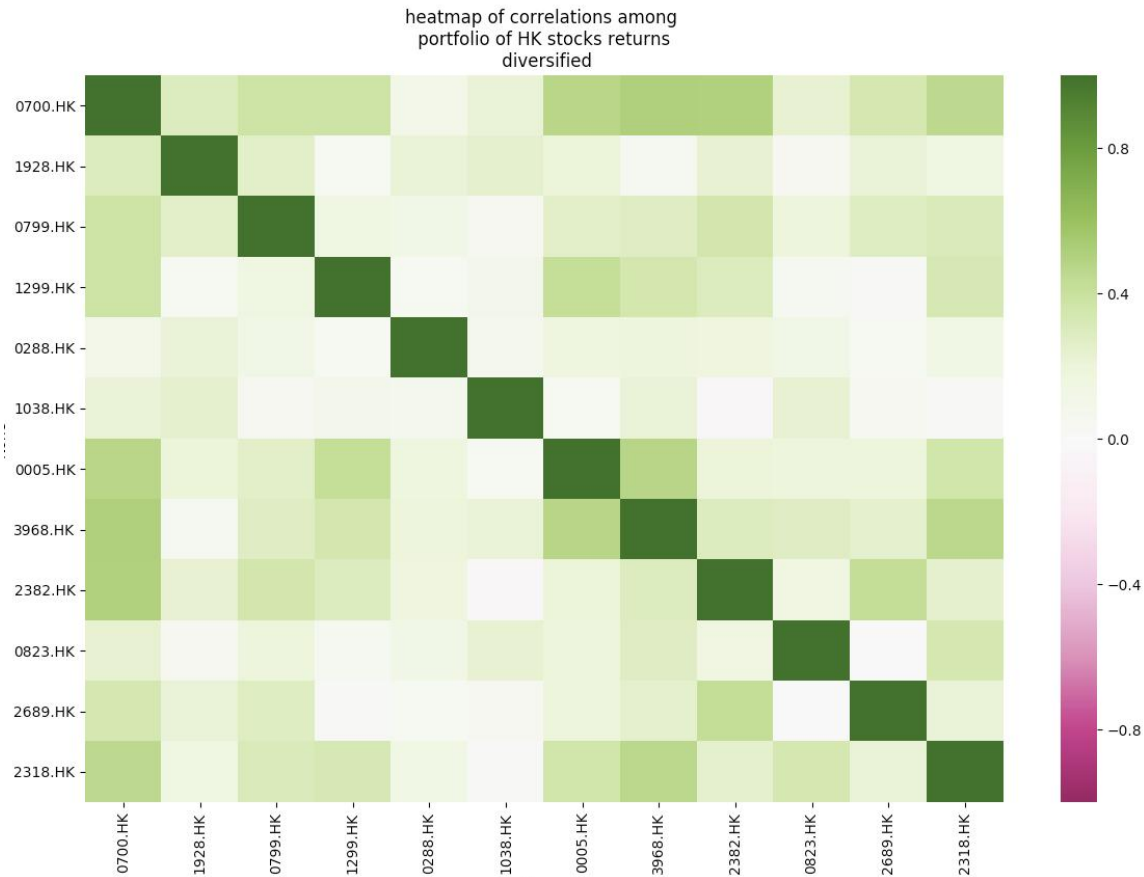
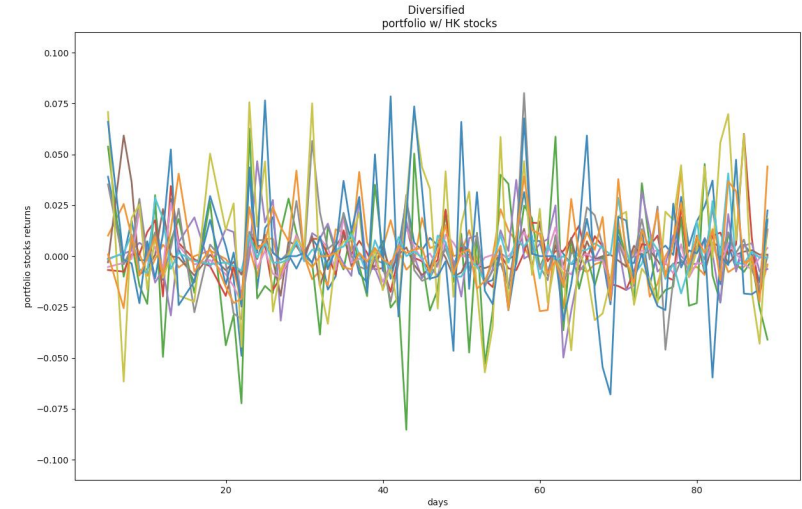


Fig. 1 Heatmap of correlations among portfolio assets' returns - Diversified (beecu002)

Portfolio Stocks Returns



Diversified Stocks

- Remarks
 - Both plot shows the trader's portfolio being not as diversified.

FDTScore: Soft-unique Analysis – ‘beecu002’,^{HK Stock}

Fig. 3 Number of assets explaining portfolio variance

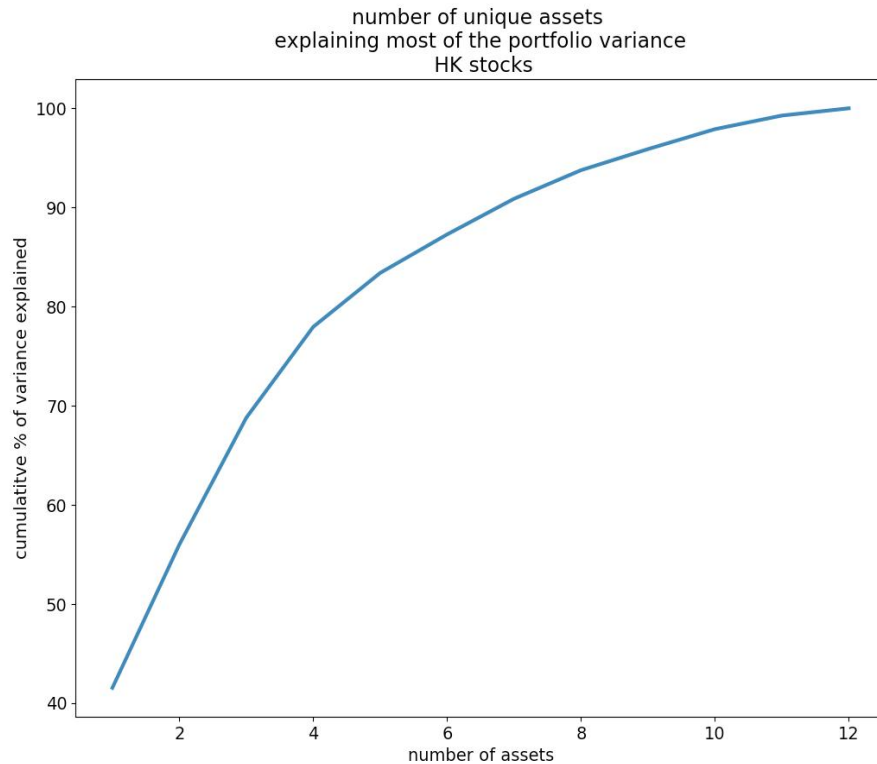
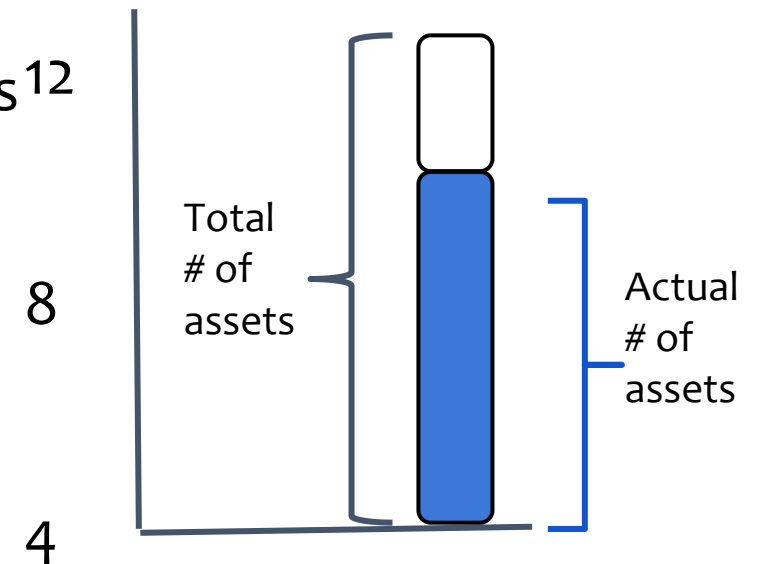
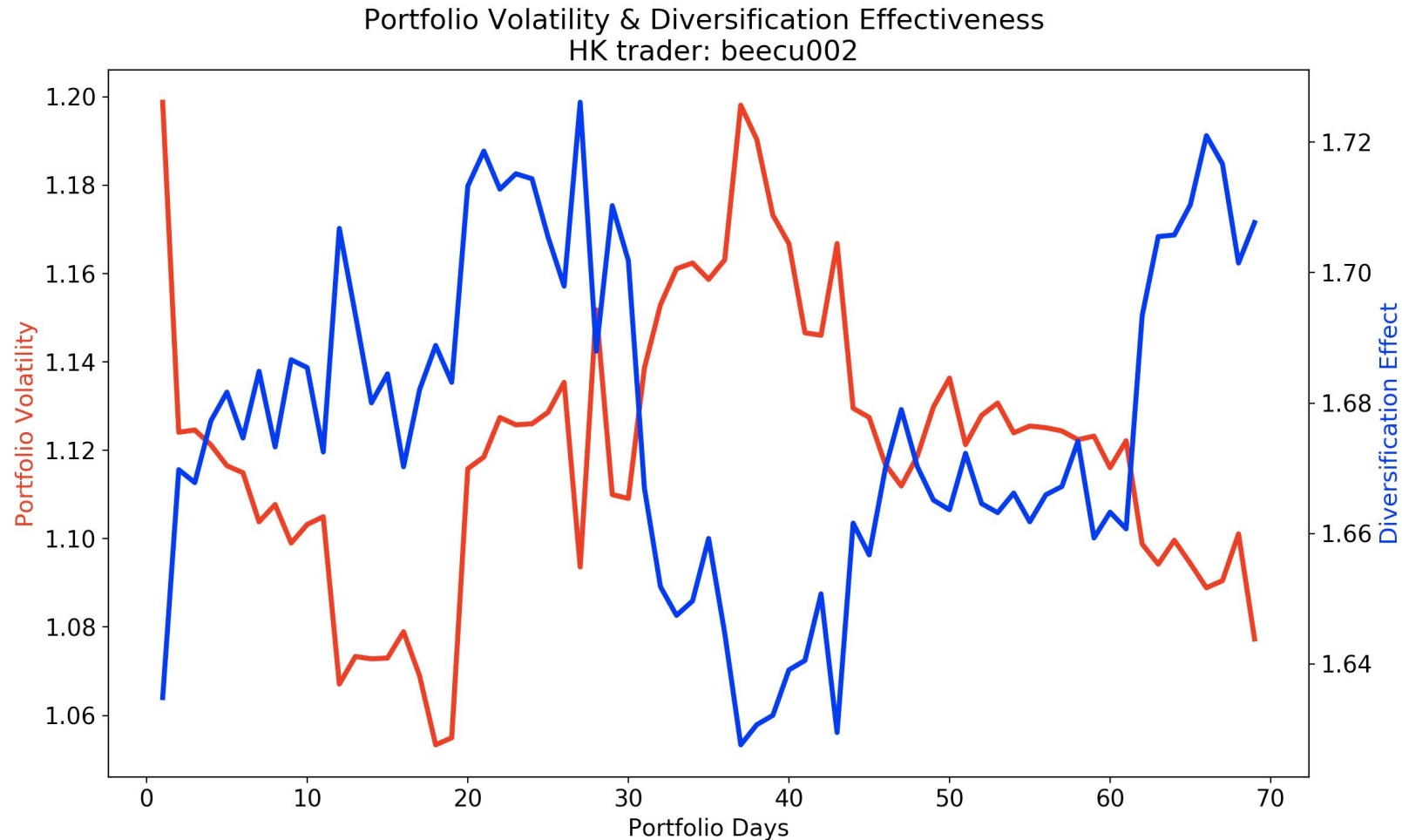


Fig. 4 Actual number of assets in portfolio

of assets¹²



FDTScore: Hedging Effect Analysis – ‘beecu002’



DE in HK market first increased then decreased and finally increased again, hence, portfolio risk was low then high and finally lowered again. Portfolio volatility range from 1 to 1.2, while DE ranges from 1.6 to 1.72 (overall portfolio risk was reduced by ~170%)

FDTScore: Diversity Analysis - "beecu003"

China A-share

FDTAI

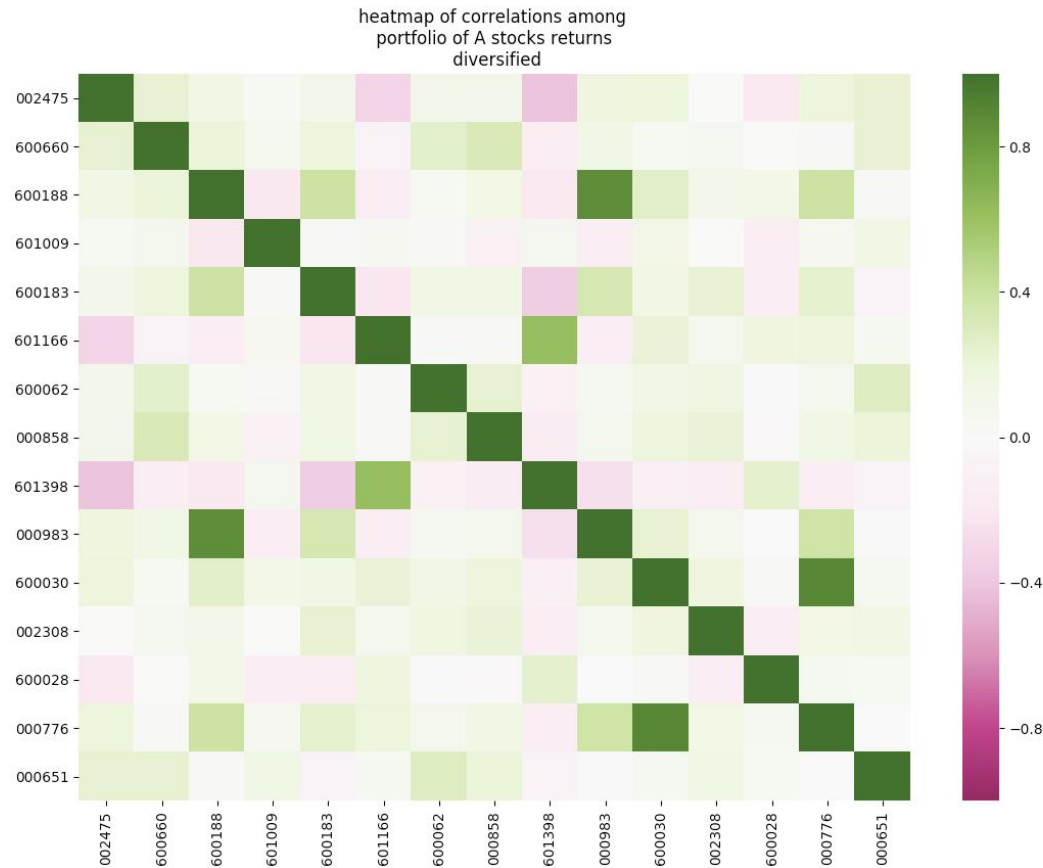
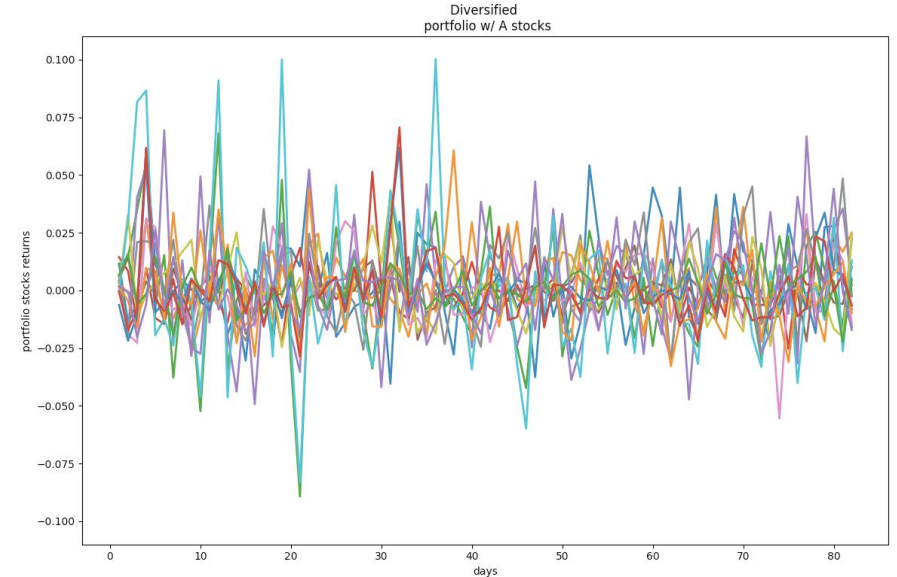


Fig. 1 Heatmap of correlations among portfolio assets' returns - Diversified (beecu002)

Portfolio Stocks Returns



Diversified Stocks

• Remarks

- The plot shows the trader's portfolio being very diversified, with some degree of hedging.

FDTScore: Soft-unique Analysis – ‘beecu003’, China A-share

Fig. 3 Number of assets explaining portfolio variance

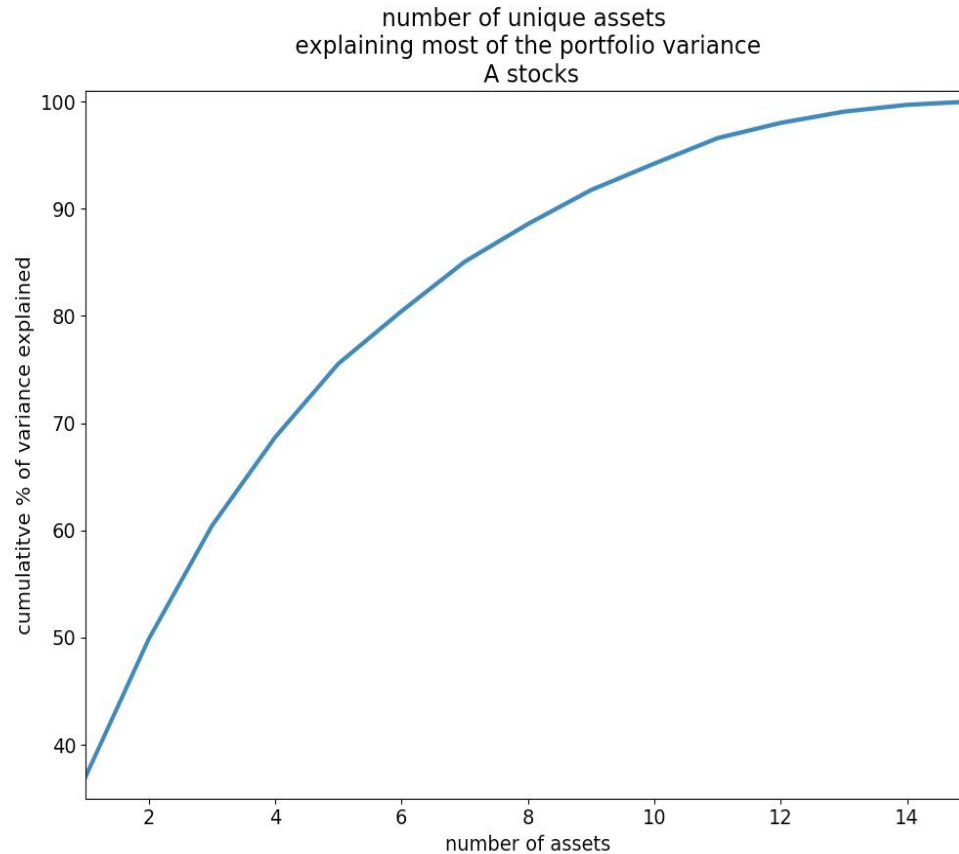
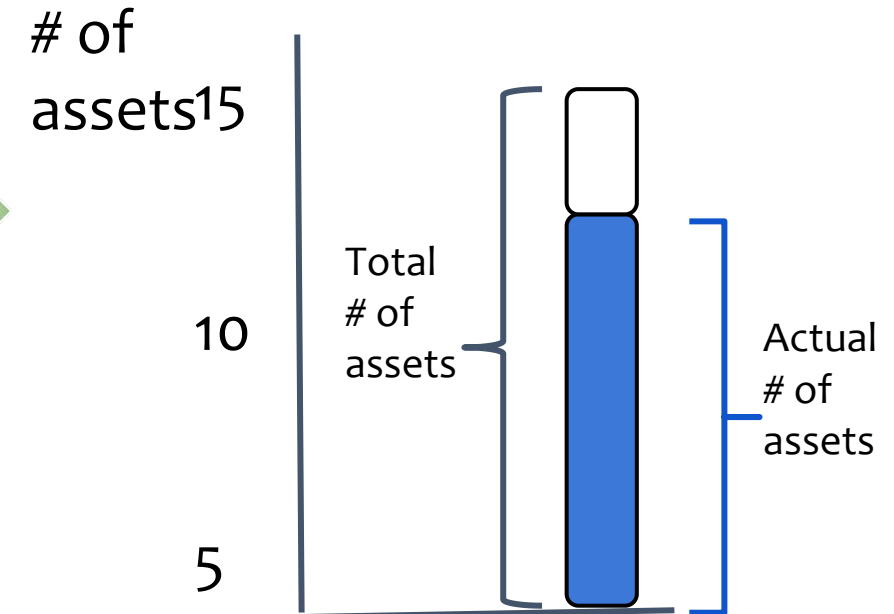
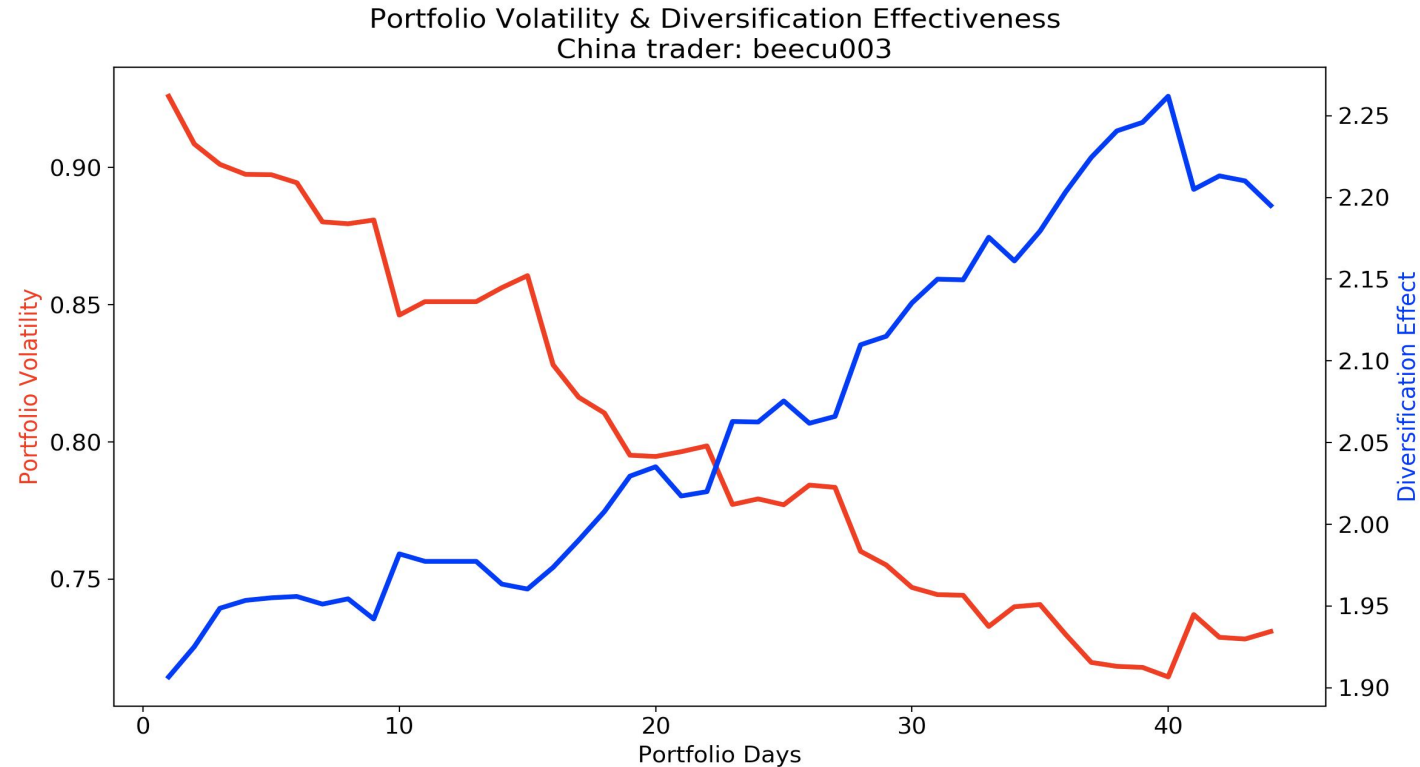


Fig. 4 Actual number of assets in portfolio



FDTScore: Hedging Effect Analysis – ‘beecu003’



Initially, when DE was lower ~1.9, portfolio volatility was higher ~0.8 - 0.9; then the trader's diversifying effect continually increased all the way to >2.2 (overall portfolio risk was reduced by 225%!), effectively reducing portfolio risk to < 0.7.

Diversity: asset similarity analysis

- Summary

1. The trader in A stocks ('beecu003') market is the most-diversified, even showing strategies of hedging.
2. The trader in US stocks ('beecu001') market is well-diversified, but not as good in A stocks market; there is less hedging.
3. The trader in HK stocks ('beecu002') market is the least-diversified, compared to the other two. As shown by the heatmap, similarities are higher among the assets.

Diversity: risk reduction effect analysis

1. Trend

The best trend (gradual and sustained improvement in DE) is seen for beecu003 in A stocks. This means the portfolio was well adjusted for increasing diversification over time such that overall risk keeps decreasing. beecu001 (US stocks) also show this trend. **This means their portfolio asset choices and weightings are chosen correctly over time.**

In contrast, beecu002 for HK stocks, DE was more random and overall risk was volatile throughout. **This means the trader should to select his asset choices and weights better.**

2. Risk and diversifying effect

Beecu003 in A stocks reduced risk *the most*, by 225%, and risk ranged from 0.70-0.95. Beecu001 in US stocks reduced risk by 195%, and risk ranged from 0.56-0.70. Beecu002 in HK stocks reduced risk *the least*, by 170%, and risk ranged from 1-1.2. This overall risk is lowest in US market and highest in HK market.